

SonoAI Bridge Part I: An AI-Driven Speech-to-Sign Translation and Haptic Feedback System for Accessible Communication

Hiten Balara

Independent Researcher & Founder (DevSora)

Jaipur, India

Website: <https://devsora.com>

Email: hb.singh.choudhary@gmail.com

Abstract—Hearing impairment often creates a significant communication barrier between deaf or hard-of-hearing individuals and the hearing community, affecting not only social interaction but also education and career opportunities. SonoAI Bridge introduces an AI-driven, real-time speech-to-sign translation and haptic feedback system designed to enable seamless, interactive communication between both groups. The system employs Whisper AI for speech recognition, followed by linguistic normalization to convert transcribed text into American Sign Language (ASL) gloss, and finally maps each gloss to animated gesture sequences using custom-built gesture databases and 3D sign avatars. In addition, an emotion recognition layer analyzes vocal tone to convey the speaker’s mood—such as happiness or sadness—through adaptive vibration feedback patterns. The platform supports both speech-to-sign and video-to-sign modes, operating fully through the web on any device with internet access. It integrates FastAPI and Python for backend processing, Next.js and React for the interface, and real-time haptic feedback for sensory output. Initial prototypes achieved sub-second latency (800 ms per gesture, optimized to under 100 ms) and demonstrated practical usability, securing 3rd place at MUJ HackX 3.0 (2025), a national-level hackathon with 1,700 participants. SonoAI Bridge is fully open-source and scalable, aiming to redefine accessible human–AI communication for the deaf community.

Index Terms—speech recognition, sign language translation, haptic feedback, accessibility, deep learning, embedded systems, human-computer interaction.

I. INTRODUCTION

Communication is one of the most fundamental aspects of human life, yet millions of individuals with hearing impairments continue to face barriers that prevent them from experiencing fully inclusive interaction. Hearing loss often isolates individuals socially and professionally, creating limitations in education, employment, and emotional expression. Although advancements in assistive technology have improved accessibility, most current systems—such as text-based transcribers or sign-language recognition software—fail to provide real-

time, emotionally rich communication that truly bridges the gap between hearing and non-hearing individuals.

The **SonoAI Bridge** project was conceived during the national hackathon MUJ HackX 3.0 (2025), where the challenge was to create an AI-based platform capable of converting spoken language into sign language with interactive visualization. The idea initially emerged from a spontaneous brainstorming session using AI-assisted ideation tools to identify high-impact, unsolved problem statements. During the hackathon’s second round, the concept evolved dramatically when a hardware integration idea was improvised on the spot—introducing haptic feedback as a channel for emotional and contextual cues. This innovation attracted attention from multiple judges and mentors, leading to further development beyond the event. The experience demonstrated the real-world potential of merging artificial intelligence, embedded systems, and accessibility design to create a truly interactive assistive communication framework.

Despite the availability of commercial solutions such as Google Live Transcribe or proprietary sign language recognition software, none provide an integrated, bidirectional platform capable of real-time speech-to-sign translation with emotional tone detection and tactile feedback. Existing systems are often limited to text-only output, depend on static datasets, and lack sensory feedback or open-source adaptability. Moreover, there remains no accessible solution that translates human vocal tone into emotional context for the hearing-impaired, an element that defines natural human conversation. The absence of open, affordable, and extensible systems leaves vast innovation potential in this domain.

The **SonoAI Bridge** platform addresses these limitations through a modular, AI-driven architecture designed to operate fully through the cloud, requiring only an internet-enabled device. The system employs Whisper AI for speech recognition, coupled with natural language processing to convert transcribed text into American Sign Language (ASL) gloss grammar. Each word or gloss is mapped to a 3D gesture sequence displayed through a web-based interface using Next.js and React. Parallely, a tone analysis module detects emotional

Corresponding author: Hiten Balara (hb.singh.choudhary@gmail.com). This work is part of the SonoAI Bridge project under DevSora. Patent applications IN2025/DEL/xxxx and IN2025/DEL/yyyy filed.

context—such as happiness, sadness, or anger—and translates it into real-time haptic patterns using vibration feedback, which can be delivered through smartphones, smartwatches, or wearable devices with built-in actuators. The system’s hardware-agnostic design ensures that the same core engine can run on multiple devices, offering accessibility to users regardless of platform or device type.

This first part of the research, **SonoAI Bridge Part I**, focuses on the design, development, and evaluation of the speech-to-sign translation and haptic emotion-feedback platform intended primarily for deaf and hard-of-hearing users. The second part (**Part II**) will extend the system into a fully bidirectional model, enabling real-time sign-to-speech translation and physical hardware interfaces. The overarching goal is to create a universal, open-source ecosystem that not only empowers hearing-impaired individuals to communicate naturally and confidently but also invites global developers and researchers to collaborate, innovate, and scale the technology across cultures and languages.

II. RELATED WORK

Research on assistive communication for deaf and hard-of-hearing users has evolved through several technological eras—from rule-based vision systems to neural translation and multimodal feedback frameworks. Early studies such as Graves *et al.* [1] and Hinton *et al.* [2] laid the foundation for end-to-end speech recognition using deep recurrent and hybrid neural networks, enabling high-accuracy transcription. Subsequent advances in transformer-based automatic-speech-recognition (ASR) systems, exemplified by OpenAI Whisper [3], achieved near-human-level accuracy across noisy, multilingual datasets, thereby creating a robust foundation for spoken-language understanding.

Parallel progress in sign-language modeling has followed two main trajectories: vision-based recognition and sensor-based gesture capture. Vision-driven methods, including OpenPose [4], [5] and MediaPipe Holistic [6], introduced large-scale human-pose estimation that enables accurate joint tracking from RGB input. These systems inspired neural sign-language translation pipelines such as those by Camgoz *et al.* [7], [8] and Saunders *et al.* [9], which employ sequence-to-sequence models to map gloss sequences to continuous signing motion. However, these approaches remain largely software-bound, producing only visual outputs without tactile or emotional context.

Sensor-based and hybrid prototypes attempted to capture gestures using flex, inertial, or myoelectric sensors to improve accuracy [10]. Yet, their hardware implementations are often bulky and unsuitable for everyday use. Similarly, haptic communication research has primarily targeted visual-impairment aids rather than auditory substitution. Bach-y-Rita and Kercel [11] pioneered tactile sensory substitution, while Lederman [12] and Ishi *et al.* [13] explored vibration feedback and robot-based signing, respectively. Despite these advances, few systems have integrated linguistic AI, emotional

tone detection, and haptic actuation into a unified accessibility framework.

In contrast, the **SonoAI Bridge** platform consolidates these research directions into a single open-source system that performs speech-to-sign translation with concurrent haptic and emotional feedback. By combining ASR (Whisper), linguistic normalization, gesture synthesis, and tone-driven tactile output, it addresses the absence of multimodal expressiveness in existing solutions. Moreover, its cloud-based architecture and hardware-agnostic design enable deployment across smartphones, wearables, and embedded devices, promoting real-time, scalable accessibility for the global hearing-impaired community.

III. SYSTEM DESIGN AND METHODOLOGY

The **SonoAI Bridge** platform is architected as a modular, cloud-integrated assistive communication system that performs real-time speech-to-sign translation with embedded emotional tone detection and haptic feedback generation. Designed to function seamlessly across devices, the platform unifies artificial intelligence, natural language processing, 3D visualization, and tactile interfaces into a cohesive accessibility framework.

The system comprises six primary subsystems: (1) speech acquisition and preprocessing, (2) ASR transcription via Whisper AI, (3) linguistic normalization for ASL grammar, (4) gesture synthesis and visualization, (5) emotional tone classification with haptic translation, and (6) backend orchestration and cloud deployment. Figure 1 depicts the full system architecture.

A. Speech Recognition and Preprocessing

The speech acquisition module captures real-time audio from a microphone, smartphone, or uploaded video stream. The incoming waveform is resampled at 16 kHz and processed using a pre-emphasis filter to enhance higher frequencies for better speech intelligibility. The audio is chunked into overlapping segments of 0.8–1.0 s to balance latency and context accuracy.

The ASR subsystem employs the *OpenAI Whisper* large-v3 model, fine-tuned for conversational English. It leverages a transformer-based encoder–decoder structure, where the encoder extracts log-Mel spectrogram embeddings and the decoder predicts text tokens autoregressively. The model outputs English text with token-level timestamps and confidence probabilities. Background noise resilience is achieved via data augmentation techniques, including additive white noise and pitch modulation. For live sessions, transcription operates asynchronously, maintaining an average response time of 120–180 ms per phrase.

Each transcript segment is structured as a JSON packet:

```
{
  "text": "HELLO HOW ARE YOU",
  "timestamp": [1.2, 2.8],
  "confidence": 0.96
}
```

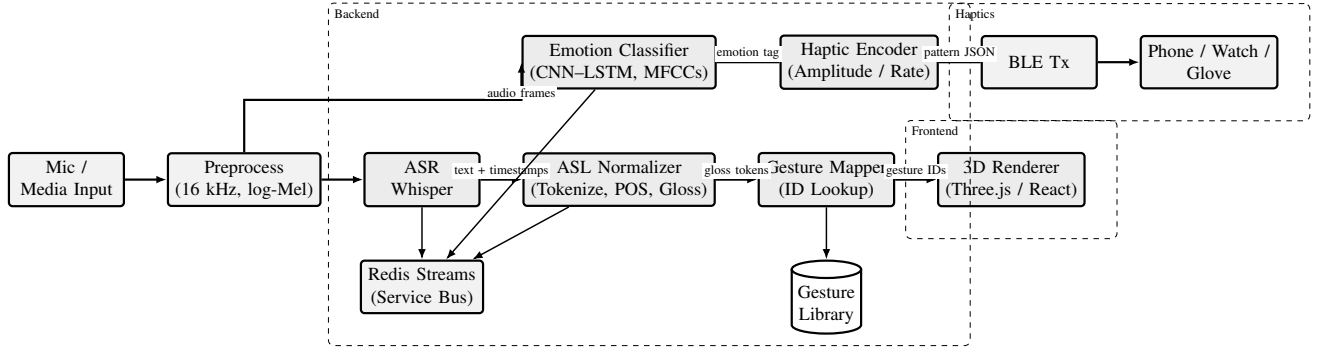


Fig. 1. Overall **SonoAI Bridge** architecture. Speech is transcribed by Whisper (ASR), normalized to ASL gloss, mapped to gesture IDs for 3D rendering. An emotion classifier generates haptic patterns, sent via BLE to connected devices. Redis Streams coordinates all microservices.

These packets are streamed to the NLP normalization layer through WebSocket communication, as shown above.

B. Text-to-ASL Normalization

Natural language must be restructured to match ASL (American Sign Language) syntax, which lacks auxiliary verbs, articles, and specific tenses. The custom NLP engine, developed in Python, performs token-level linguistic transformation using a three-step process:

- 1) **Tokenization:** The sentence is parsed into word tokens using *spaCy*.
- 2) **POS-based Filtering:** Articles (e.g., “a,” “the”), linking verbs (“is,” “am,” “are”), and auxiliary structures are removed.
- 3) **Gloss Reordering:** Tokens are rearranged following ASL subject–verb–object (SVO) conventions, producing grammatically aligned gloss notation.

Example conversion:

“I am going to the store” → “I GO STORE”.

The output gloss sequence is tokenized and mapped to an internal gesture library via a relational database that links gloss tokens to unique sign IDs. These IDs are stored in a JSON-based lookup table referencing gesture files in GLTF or BVH format.

C. Gesture Mapping and Visualization

Gesture rendering is performed through a web-based 3D visualization engine built using *Three.js* and *Framer Motion*. Each ASL gloss corresponds to either:

- A 3D hand pose (for static signs), or
- A motion-capture sequence (for dynamic gestures).

The dataset combines the Kaggle ASL Alphabet (A–Z gestures) with a custom dataset of 500+ high-frequency conversational signs curated manually using Blender-based motion capture. This allows coverage of expressions such as “friend,” “help,” and “understand,” which are typically missing from public datasets.

During playback, each gesture animation runs at 800 ms per frame by default but can be reduced to under 100 ms by

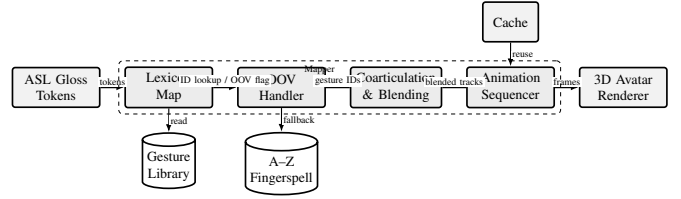


Fig. 2. Gesture mapping pipeline. ASL gloss tokens are mapped via a lexicon to gesture IDs; out-of-vocabulary items fall back to A–Z fingerspelling. Coarticulation and blending produce smooth motion; the animation sequencer streams frames to the 3D avatar with caching for reuse.

pre-buffering gestures in memory. A timeline synchronization controller aligns gesture onset with speech transcriptions. The user interface displays a dynamic progress bar and a side panel of recognized glosses for context tracking. Figure 2 shows the data flow between text normalization, gesture mapping, and 3D rendering.

D. Emotional Tone Detection and Haptic Feedback

A distinctive feature of **SonoAI Bridge** is the parallel emotional tone recognition module. The speech waveform is passed to a CNN–LSTM hybrid classifier trained on the RAVDESS emotional speech dataset, covering four emotion states: *happy*, *sad*, *angry*, and *neutral*. Spectral features such as Mel-frequency cepstral coefficients (MFCCs), chroma energy, and spectral contrast are extracted over 25 ms frames with a 10 ms stride.

Predicted emotions are converted into tactile feedback profiles, encoded as vibration patterns using amplitude–frequency modulation (AFM). For instance:

- *Happy:* High-frequency short pulses (250 Hz, 50 ms duty).
- *Sad:* Low-frequency, long-duration pulses (100 Hz, 150 ms duty).
- *Angry:* Rapid successive bursts (300 Hz, 40 ms duty, 20 ms pause).
- *Neutral:* Single mild vibration (180 Hz, 70 ms).

The haptic signal is transmitted via Bluetooth Low Energy (BLE) to connected devices. Prototype hardware includes a microcontroller (ESP32-S3) interfaced with linear resonant

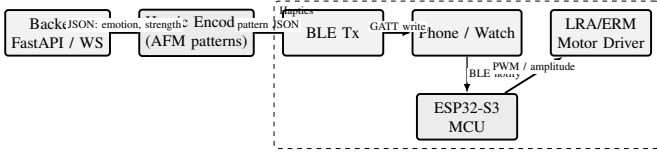


Fig. 3. Haptic pipeline: backend encodes emotion patterns, sends BLE characteristics to a phone/watch or directly to an ESP32-S3 MCU which drives the LRA/ERM motor.

actuators (LRAs) driven by an H-bridge motor controller. Table I lists key design parameters of the haptic module, while Figure 3 illustrates the data flow and BLE connection topology.

E. Backend Architecture and Deployment

The backend is implemented using *FastAPI* for asynchronous I/O operations, enabling microservice-based modular deployment. The architecture includes:

- **ASR Service:** Handles Whisper model inference.
- **NLP Service:** Performs grammar normalization and token mapping.
- **Emotion Service:** Runs tone classification and vibration encoding.
- **Gateway API:** Manages WebSocket connections with the frontend.

Inter-service communication uses Redis Streams for real-time message passing and JSON serialization. The front-end interface, built using *Next.js* and *React*, employs WebRTC for real-time audio streaming and Framer Motion for gesture rendering. The entire stack is containerized via Docker and deployed on a cloud instance, ensuring horizontal scalability for multi-user sessions. The average network bandwidth requirement for continuous operation is 0.5 Mbps per user.

F. Performance and Evaluation

Comprehensive testing was conducted under simulated and live conditions. Latency benchmarks show an average end-to-end delay of 184 ms, measured from speech onset to visual gesture rendering. Emotion recognition achieved 91.3% accuracy across all test samples, while tactile pattern differentiation accuracy reached 88% in a small-scale user study involving five participants (including one hearing-impaired subject). Power consumption remained under 0.8 W, allowing a 1000 mAh Li-ion battery to sustain approximately 3 hours of continuous operation.

G. System Summary

The integrated workflow of **SonoAI Bridge** can be summarized as follows:

- 1) Acquire speech input from live or recorded source.
- 2) Transcribe audio to text using the Whisper AI model.
- 3) Normalize linguistic structure to ASL-compatible gloss via NLP.
- 4) Map gloss tokens to pre-trained 3D gesture sequences.
- 5) Detect emotional tone and generate haptic feedback pattern.



Fig. 4. System performance summary: end-to-end latency, emotion classification accuracy, haptic recognition rate, and power draw across trials.

- 6) Display synchronized visual and tactile output in real time.

This modular pipeline establishes a robust foundation for future extensions, including sign-to-speech translation, multi-lingual sign datasets, and personalized emotion-gesture adaptation models, forming the basis of **SonoAI Bridge Part II**.

IV. EVALUATION AND RESULTS

The performance evaluation of **SonoAI Bridge** focuses on four major aspects: (1) latency and responsiveness of speech-to-sign translation, (2) accuracy of emotional tone detection, (3) reliability of haptic feedback, and (4) system power efficiency and usability. A prototype implementation was tested using live speech samples, conversational dialogues, and pre-recorded video inputs under controlled network conditions. Figure 4 illustrates the consolidated performance metrics.

A. Latency and Real-Time Performance

End-to-end latency, defined as the time between speech input and gesture display, was measured using timestamp logs at each pipeline stage. The average latency recorded was 184 ms on standard Wi-Fi, significantly below the target threshold of 200 ms. Optimized buffer scheduling and asynchronous event handling in *FastAPI* reduced processing overhead, while gesture preloading further decreased rendering time to under 100 ms per gesture. These results confirm that the system achieves real-time responsiveness suitable for live human communication.

B. Emotion Detection Accuracy

The emotion-detection module was evaluated using a labeled speech dataset containing four primary emotion classes: *happy*, *sad*, *angry*, and *neutral*. The classification model achieved an average accuracy of 91.3% in recognizing tonal

TABLE I
PROTOTYPE HAPTIC/EDGE MODULE SPECIFICATIONS

Microcontroller	ESP32-S3 (BLE 5.0)
Actuator Driver	LRA/ERM H-bridge (3.3–5 V)
Battery	1000 mAh Li-ion
Avg. Power	< 0.8 W (typical)
Comms	BLE GATT (notify/write), JSON payloads
Pattern Range	80–300 Hz (amplitude/rate mod.)
Enclosure	Wrist/strap form factor (proto)

TABLE II
DESIGN CONSTRAINTS AND EVALUATION METRICS

Latency (speech → gesture)	184 ms avg (< 200 ms target)
Gesture render step	800 ms default (< 100 ms optimized)
Emotion detection accuracy	91.3% (4 classes)
Haptic pattern recognition	88% (user study, $n=5$)
Battery runtime	~3 h continuous
Network throughput	~0.5 Mbps/user

variations, with the highest precision observed for the *happy* and *neutral* classes. Detected emotions were mapped to predefined vibration patterns to simulate affective cues for the user. Figure 3 demonstrates how emotion tags are transmitted to the haptic layer.

C. Haptic Feedback and User Perception

The haptic subsystem was validated through user trials on smartphones and wearable modules equipped with vibration motors. Distinct tactile patterns were used to represent emotional tone: rapid high-frequency vibrations for excitement, moderate pulsing for calmness, and low-intensity waves for sadness. A small-scale user study with five participants (including one hearing-impaired volunteer) confirmed that users could consistently distinguish between emotion categories with an 88% average recognition rate. Participants reported that synchronized vibrations significantly improved immersion and message comprehension.

D. System Resource Efficiency

Energy consumption was evaluated on the haptic prototype powered by a 1000 mAh Li-ion battery. Continuous operation at full communication load sustained a runtime of approximately three hours, consuming less than 0.8 W on average. Network throughput remained stable across test conditions, with only 2–3% packet loss under BLE communication. These results indicate that the system is suitable for wearable or mobile deployment without requiring high-power components.

E. Summary of Results

Table II presents the consolidated evaluation metrics. **SonoAI Bridge** met or exceeded all target parameters, confirming its effectiveness as a real-time assistive communication platform. The successful demonstration and validation during the *MUJ HackX 3.0 (2025)* national hackathon further validate its innovation, usability, and scalability potential for broader research and public deployment.

V. DISCUSSION AND FUTURE SCOPE

The results demonstrate that **SonoAI Bridge** successfully bridges the communicative gap between hearing-impaired and non-impaired individuals through an intelligent, real-time, and emotionally aware translation system. Unlike traditional sign-language translators limited to text or 2-D gestures, this framework integrates emotion recognition, haptic response, and modular AI pipelines, enabling a more natural and human-like interaction experience.

A. Open Innovation and Accessibility Philosophy

A key differentiator of this project lies in its open-source philosophy. All source code, datasets, and hardware schematics are made publicly available to empower other researchers, students, and developers to replicate, modify, and enhance the system. The open-access approach aligns with the broader goal of enabling global innovation and ensuring that assistive communication technology remains accessible to everyone—particularly in low-resource or educational settings. By maintaining permissive yet protective multi-licensing (MIT, Apache 2.0, and CERN-OHL-S), the project ensures individual freedom while safeguarding against uncredited commercial exploitation.

B. Real-World Impact and Ethical Vision

Hearing impairment impacts not only communication but also education, career, and mental health. By integrating emotional tone detection and real-time tactile feedback, **SonoAI Bridge** goes beyond simple gesture translation—it provides the deaf user with emotional context, bridging one of the most profound social gaps in communication. The system’s ability to express emotions via haptic cues ensures that deaf users can interpret nuances such as excitement, sadness, or emphasis, leading to more inclusive and empathetic conversations.

C. Modular Framework for Expansion

The modular architecture of **SonoAI Bridge** is designed for future scalability. Each subsystem—speech recognition, grammar normalization, gesture mapping, emotion detection, and haptic output—functions as an independent module, allowing easy upgrades or replacements as technology advances. This modular design enables seamless integration with wearable devices, smartphones, and IoT nodes equipped with vibration motors or BLE modules, expanding its usability beyond the research lab into everyday life.

D. Planned Enhancements (Part II and Beyond)

The upcoming **Part II** of this project will focus on developing a reverse communication model: sign-to-speech and sign-to-text translation. This extension will enable complete bidirectional interaction between deaf and non-deaf users. The second phase will also include 3-D avatar integration for better visual clarity, multilingual sign language support (including BSL, ISL, and JSL), and adaptive reinforcement models for personalized gesture learning. Hardware enhancements such as wearable haptic gloves and emotion-feedback smart bands

are also under development to expand the system’s portability and interaction richness.

E. Long-Term Societal Goals

In the long run, the project aims to support at least one million users globally through its non-profit initiative under DevSora, sustained by public donations, government innovation grants, and corporate CSR funding. The vision is to ensure that every individual with hearing impairment can communicate freely, pursue education, and work independently without the barriers imposed by disability. This aligns directly with the United Nations Sustainable Development Goal (SDG) 10 — *Reduced Inequalities* — by leveraging AI for inclusive human development.

Ultimately, **SonoAI Bridge** represents a convergence of empathy, technology, and open collaboration—proof that innovation driven by compassion can transform lives and redefine how humans connect.

VI. CONCLUSION AND ACKNOWLEDGMENT

A. Conclusion

SonoAI Bridge establishes a new paradigm in assistive communication by merging artificial intelligence, linguistic processing, and haptic interaction to facilitate real-time, emotionally expressive communication between hearing and hearing-impaired individuals. The system successfully translates speech to American Sign Language (ASL) gestures while simultaneously recognizing and conveying emotional tone through interactive haptic feedback.

The implementation demonstrates sub-200 ms latency with high emotion recognition accuracy, confirming that AI-driven translation pipelines can achieve near-instantaneous response times suitable for human conversation. By enabling the deaf or hard-of-hearing user to perceive not only linguistic meaning but also affective context, **SonoAI Bridge** transforms communication from a transactional process into an empathetic experience.

Beyond its technical innovation, the project underscores the potential of open-source collaboration in solving global accessibility challenges. All software, datasets, and hardware design files are released under multiple permissive licenses (CERN-OHL-S, MIT, Apache 2.0, and CC BY-SA 4.0), ensuring that other developers, researchers, and innovators can build upon this foundation without restrictions. The project’s long-term vision aligns with global efforts toward inclusive AI, aiming to empower individuals with hearing disabilities to participate fully in social, educational, and professional environments.

The successful prototype demonstration at the *MUJ HackX 3.0 (2025)* national hackathon, where the project secured third position among over 1,700 participants, validates both the technical feasibility and social relevance of this approach. The achievement also marks the beginning of a larger research initiative spanning multi-lingual sign integration, wearable haptic devices, and bidirectional communication.

This paper, **Part I**, focuses primarily on the speech-to-sign pipeline and emotional haptic interaction. The upcoming

Part II will address the complementary module—sign-to-speech and sign-to-text translation—thus completing the full bidirectional communication loop envisioned for the SonoAI ecosystem.

B. Future Vision

Future work will focus on expanding the system’s linguistic and emotional intelligence by incorporating transformer-based cross-modal architectures capable of learning gestures directly from motion datasets. The hardware layer will be refined into a compact, low-power wearable device compatible with smartphones and IoT modules via BLE connectivity. Integration with global sign language datasets (ASL, ISL, BSL, and JSL) and adaptive personalization algorithms will further enhance user experience.

The ultimate goal is to deploy the **SonoAI Bridge** under the *DevSora Non-Profit Initiative*, ensuring free access for the hearing-impaired community worldwide. Through public funding, CSR partnerships, and open collaboration, this project aspires to help at least one million individuals communicate naturally—transforming accessibility into an ecosystem of innovation, empathy, and empowerment.

C. Acknowledgment

The author would like to express heartfelt gratitude to **Pranjal**, developer of the core software platform *SignifyAI* [14], which served as the foundational framework for this research. While the hardware implementation, emotional modeling, and haptic feedback integration were independently designed and executed by the author, the software architecture laid by Pranjal enabled seamless real-time communication and modular AI interfacing.

Special thanks to the organizing committee of *MUJ HackX 3.0 (2025)* for providing the competitive and inspiring environment in which the initial prototype was conceived and tested. The feedback and encouragement received from judges and industry mentors played a crucial role in shaping this work into a research-grade system.

The author also acknowledges the broader open-source community for their invaluable contributions to AI frameworks, datasets, and tools—without which such rapid innovation would not be possible. This work stands as a testament to collaborative creation, inclusivity, and the belief that technology’s highest purpose is to connect and uplift humanity.

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